

Computability & Logic, Exercises to Lectures 21+22

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Theoretical Exercises

From John C. Martin book

Recommendation:

- Minimal set of recommended exercises: 9.3, 9.8, 9.9, 9.12(a,b), 9.27, 9.32(a,b)
- Optional exercises: 9.12(c), 9.15, 9.29, 9.32(c), 9.34

Problem Reductions

- 9.3 (discussed at lecture)
- 9.8
- 9.9 **Extra question:** Conclude that $Accepts\{-\Lambda\}$ is undecidable.

Decidability Questions

- 9.12 (a,b) (c is also interesting)
- 9.15 (optional, only if sufficient time)

Post-Correspondence Problem

- 9.27
- 9.29 (optional, only if sufficient time)

Decision Problems on Context-Free Languages

- 9.32 (Hint: consider reductions involving $CFGGeneratesAll$, see Thm 9.23)
- 9.34